EGM^n: The Sequential Endogenous Grid Method

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Abstract

Heterogeneous agent models with multiple decisions (consumption, labor, portfolio allocation) face a computational dilemma. Joint optimization over all choices is slow because it requires evaluating high-dimensional optimization problems at every point in a large state space. Imposing strong separability restrictions speeds things up but rules out economically interesting interactions between decisions. We show how to resolve this tension through two complementary contributions. First, we recognize that while decisions are economically simultaneous (made under the same information set), they need not be computationally joint. By decomposing the problem into sequential stages, each stage exploits the Endogenous Grid Method's efficiency, chaining speed gains across decisions. Second, the resulting endogenous grids are warped or unstructured, creating an interpolation challenge that has limited prior sequential approaches. We develop interpolation methods using both curvilinear techniques and Gaussian Process Regression that overcome this bottleneck while preserving accuracy. The method solves multidecision heterogeneous agent models in a fraction of the time required by standard grid search approaches.

Keywords endogenous grid method, dynamic programming, machine learning

1 Introduction

Solving heterogeneous agent models with multiple decisions has traditionally required either nested optimization (which is slow) or restrictive assumptions about preferences (which limits economic relevance). The Endogenous Grid Method revolutionized consumption-savings problems by inverting first-order conditions, but extending this insight to problems with multiple continuous choices and high-dimensional state spaces has remained an open challenge. We show how to decompose such problems into sequential stages that each admit an EGM step, effectively chaining the efficiency gains across decisions.

1.1 Background and Literature

Macroeconomic modeling aims to describe a complex world of agents interacting with each other and making decisions in a dynamic setting. The models are often very complex, require strong underlying assumptions, and use a lot of computational power to solve. One of the most common methods to solve these complex problems is using a grid search method to solve the model. The Endogenous Grid Method (EGM) developed by Carroll [2006] allows dynamic optimization problems to be solved in a more computationally efficient and faster manner than the previous method of convex optimization using grid search. Many problems that before took hours to compute became much easier to solve and allowed macroeconomists and computational economists to focus on estimation and simulation. However, the Endogenous Grid Method is limited to a few specific classes of problems. Recently, the classes of problems to which EGM can be applied have been expanded (see, e.g., Barillas and Fernández-Villaverde [2007], Maliar and Maliar [2013], Fella [2014], White [2015], Iskhakov et al. [2017]), but with every new method comes a new set of limitations.

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This paper introduces a new approach to EGM in a multivariate setting. We call the method Sequential EGM (or EGM^n) and show how to break down complex problems into a sequence of simpler, smaller, and more tractable problems, along with developing multidimensional interpolation methods that make this practical. The key distinction between EGM^n and nested approaches lies in how we treat the computational structure. Nested EGM still embeds a grid search or root-finding step within an outer EGM step. Sequential EGM chains multiple EGM inversions, exploiting separability to avoid optimization entirely when applicable. This requires stronger structural assumptions (separable utility or independent transitions), but the payoff is elimination of costly inner-loop optimization and generation of interpretable intermediate value functions at each stage. The method reveals the problem's natural information flow: which decisions shed state variables, which create new post-decision states, and how to order computations to minimize dimensionality at each step.

Carroll [2006] first introduced the Endogenous Grid Method as a way to speed up the solution of dynamic stochastic consumption-savings problems. The method consists of starting with an exogenous grid of post-decision states and using the inverse of the first-order condition to find the optimal consumption policy that rationalizes such post-decision states. Given the optimal policy and post-decision states, it is straightforward to calculate the initial pre-decision state that leads to the optimal policy. Although this method is certainly innovative, it only applied to a model with one control variable and one state variable. Barillas and Fernández-Villaverde [2007] further extend this method by including more than one control variable in the form of a labor-leisure choice, as well as a second state variable for stochastic persistence.

Hintermaier and Koeniger [2010] introduce a model with collateral constraints and non-separable utility and solve using an EGM method that allows for occasionally binding constraints among endogenous variables. Jørgensen [2013] evaluates the performance of the Endogenous Grid Method against other methods for solving dynamic stochastic optimization problems and finds it to be fast and efficient. Maliar and Maliar [2013] develop the Envelope Condition Method based on a similar idea as the Endogenous Grid Method, avoiding the need for costly numerical optimization and grid search. However, their model is limited to infinite horizon problems as it is a forward solution method.

Further development into a multivariate Endogenous Grid Method expanded the ability of researchers to solve models efficiently. White [2015] formally characterized the conditions for the Endogenous Grid Method and developed an interpolation method for structured non-rectilinear, or curvilinear, grids. Iskhakov [2015] establishes conditions for multivariate EGM through triangular systems of first-order conditions, where each FOC can be solved sequentially by substituting previously determined controls. This triangular structure is less restrictive than the full separability we exploit, but requires root-finding operations that our approach eliminates when separability holds. Ludwig and Schön [2018] also develops a novel interpolating method using Delaunay triangulation of the resulting unstructured endogenous grid. However, the authors show that the gains from avoiding the grid search method are often offset by the costly construction of the triangulation.

For the papers discussed above, continuity and smoothness of the value and first-order conditions are strict requirements. Fella [2014] first introduced the Generalized Endogenous Grid Method (GEGM) to solve non-convex problems. The idea is based on evaluating necessary but not sufficient candidates for the first-order condition in overlapping regions of the state space. Arellano et al. [2016] use the Envelope Condition Method to solve a sovereign default risk model with similar efficiency gains to EGM. Iskhakov et al. [2017] further advances the methodology by using extreme errors to solve discrete choice problems with Endogenous Grid Method. These methods however were only applied to a single control variable and a single state variable. Druedahl and Jørgensen [2017] introduces the G2EGM to handle non-convex problems with more than 1 control variable and more than 1 state variable. This method is also capable of handling occasionally binding constraints which previous multivariate EGM methods were not.

Clausen and Strub [2020] formalize the applicability of the Endogenous Grid Method and its extensions to discrete choice models and discuss the nesting of problems to efficiently find accurate solutions. Druedahl [2021] similarly suggest the nesting of problems to efficiently use the Endogenous Grid Method within problems with multiple control variables. However, while these nested methods reduce the complexity of solving these models, they often still require grid search methods as is the case with Druedahl [2021].

Finally, this paper contributes to the literature of solving dynamic optimization problems using machine learning tools. Scheidegger and Bilionis [2019] introduce the use of Gaussian Process Regression to compute global solutions for high-dimensional dynamic stochastic problems. Maliar et al. [2021] use non-linear regression and neural networks to estimate systems of equations that characterize dynamic economic models.

1.2 Methodology and Contributions

The purpose of this paper is to describe a new method for solving dynamic optimization problems efficiently and accurately while avoiding convex optimization and grid search methods with the use of the Endogenous Grid Method and first-order conditions. We call the method Sequential EGM (or EGM^n) and introduce a novel way of breaking down complex problems into a sequence of simpler, smaller, and more tractable problems, along with an exploration of new multidimensional interpolation methods that can be used to solve these problems. This paper also illustrates an example of how Sequential EGM can be used to solve a dynamic optimization problem in a multivariate setting.

The sequential Endogenous Grid Method consists of three major parts: First, the problem to be solved should be broken up into a sequence of smaller problems that themselves don't add any additional state variables or introduce asynchronous dynamics with respect to the uncertainty. If the problem is broken up in such a way that uncertainty can happen in more than one period, then the solution to this sequence of problems might be different from the aggregate problem due to giving the agent additional information about the future by realizing some uncertainty. Second, we evaluate each of the smaller problems to see if they can be solved using the Endogenous Grid Method. This evaluation is of greater scope than the traditional Endogenous Grid Method, as it allows for the resulting exogenous grid to be non-regular. If the subproblem can not be solved with EGM, then convex optimization is used. Third, if the exogenous grid generated by the EGM is non-regular, then we use a multidimensional interpolation method that takes advantage of machine learning tools to generate an interpolating function. Solving each subproblem in this way, the sequential Endogenous Grid Method is capable of solving complex problems that are not solvable with the traditional Endogenous Grid Method and are difficult and time-consuming to solve with convex optimization and grid search methods.

The Sequential Endogenous Grid Method is capable of solving multivariate dynamic optimization problems in an efficient and fast manner by avoiding grid search. This should allow researchers and practitioners to solve more complex problems that were previously not easily accessible to them, but more accurately capture the dynamics of the macroeconomy. By using advancements in machine learning techniques such as Gaussian Process Regression, the Sequential Endogenous Grid Method is capable of solving problems that were not previously able to be solved using the traditional Endogenous Grid Method. In particular, the Sequential Endogenous Grid Method is different from NEGM in that it allows for using more than one Endogenous Grid Method step to solve a problem, avoiding costly grid search methods to the extent that the problem allows. Additionally, the Sequential Endogenous Grid Method often sheds light on the problem by breaking it down into a sequence of simpler problems that were not previously apparent. This is because intermediary steps in the solution process generate value and marginal value functions of different pre- and post-decision states that can be used to understand the problem better.

1.3 Outline

Section ?? presents a basic model that illustrates the sequential Endogenous Grid Method in one dimension, showing how a three-choice problem (labor, consumption, portfolio) can be solved by sequential decomposition. Section ?? introduces a more complex problem with two state variables to demonstrate how EGM^n handles genuinely high-dimensional state spaces. Section ?? addresses the interpolation challenge: EGM generates grids that are warped or unstructured, requiring specialized interpolation methods that we develop using both curvilinear techniques and Gaussian Process Regression. Section ?? provides practical guidance on when and how to apply Sequential EGM, identifying the mathematical structures (separability, invertibility) that make the method applicable. Finally, Section ?? concludes.

2 The Sequential Endogenous Grid Method

Consider the challenge facing a researcher who wants to solve a model where households simultaneously choose consumption, labor supply, and portfolio allocation. Standard approaches face a dilemma: either optimize jointly over all three choices (which requires evaluating a three-dimensional optimization at every state space point), or impose separability restrictions that may not reflect realistic preferences. We show how sequential decomposition offers a third way that preserves generality while maintaining computational efficiency.

2.1 Problem Setup

The baseline problem which we use to demonstrate the Sequential Endogenous Grid Method (EGM^n) is a discrete time version of Bodie et al. [1992] where a consumer has the ability to adjust their labor as well as their consumption in response to financial risk. The objective consists of maximizing the present discounted lifetime utility of consumption and leisure.

$$\mathrm{V}_0(B_0, \theta_0) = \max \mathbb{E}_t \left[\sum_{n=0}^{T-t} \beta^n \mathrm{u}(C_{t+n}, Z_{t+n}) \right]. \tag{1}$$

In particular, this example makes use of a utility function that is based on Example 1 in the paper, which is that of additively separable utility of labor and leisure as

$$\mathrm{u}(C, Z) = u(C) + h(Z) = \frac{C^{1-\rho}}{1-\rho} + \nu^{1-\rho} \frac{Z^{1-\zeta}}{1-\zeta} \tag{2}$$

where the term $\nu^{1-\rho}$ scales the leisure utility to have the same curvature as consumption utility, following the approach in Mertens and Ravn [2011].[1] The use of additively separable utility is ad-hoc, as it will allow for the use of multiple EGM steps in the solution process, as we'll see later. For the remainder of the analysis, we work with normalized variables (lowercase) where consumption $c = C/P$ and leisure z represent quantities relative to permanent income or in natural units.

This model represents a consumer who begins the period with a level of bank balances b_t and a given wage offer θ_t. Simultaneously, they are able to choose consumption, labor intensity, and a risky portfolio share with the objective of maximizing their utility of consumption and leisure, as well as their future wealth.

The problem can be written in normalized recursive form[2] as

$$\begin{aligned} v_t^0(b_t, \theta_t) &= \max_{\{c_t, z_t, \varsigma_t\}} \mathrm{u}(c_t, z_t) + \beta \mathbb{E}_t \left[\Gamma_{t+1}^{1-\rho} v_{t+1}^0(b_{t+1}, \theta_{t+1}) \right] \\ &\text{s.t.} \\ \ell_t &= 1 - z_t \\ m_t &= b_t + \theta_t \ell_t \\ a_t &= m_t - c_t \\ \mathbb{R}_{t+1} &= \mathsf{R} + (\mathbf{R}_{t+1} - \mathsf{R})\varsigma_t \\ b_{t+1} &= a_t \mathbb{R}_{t+1} / \Gamma_{t+1} \end{aligned} \tag{4}$$

where we impose non-negativity constraints $c_t \geq 0$, $z_t \in [0, 1]$, and $\varsigma_t \in [0, 1]$. Throughout, we assume standard constraint qualifications hold such that interior solutions satisfy first-order conditions.[3] Here, ℓ_t denotes the time supplied to labor net of leisure, m_t denotes the market resources totaling bank balances and labor income, a_t denotes the amount of saving assets held by the consumer, and ς_t denotes the risky share of assets, which induces a portfolio return $\mathbb{R}_{t+1}$ that results in next period's bank balances b_{t+1} normalized by next period's permanent income Γ_{t+1}.

[1] An alternative formulation for the utility of leisure is to state it in terms of the disutility of labor as $h(\ell) = -\zeta \frac{\ell^{1+\nu}}{1+\nu}$, which gives $h'(z) = \zeta(1-z)^\nu$ and $h'^{-1}(x) = 1 - (x/\zeta)^{1/\nu}$. Note that this formulation does not support a balanced growth path because leisure utility is not homogeneous of degree $1-\rho$ in permanent income. A BGP-consistent specification would require $h(Z, P) = (\nu P)^{1-\rho} \frac{Z^{1-\zeta}}{1-\zeta}$, making leisure utility scale with permanent income.

[2] As in Carroll [2009], where the utility of normalized consumption and leisure is defined as

$$\mathrm{u}(c_t, z_t) = P_t^{1-\rho} \frac{c_t^{1-\rho}}{1-\rho} + (\nu P_t)^{1-\rho} \frac{z_t^{1-\zeta}}{1-\zeta} \tag{3}$$

[3] Specifically, we assume: (i) utility and value functions are twice continuously differentiable in the interior of the constraint set; (ii) the Inada conditions $\lim_{c \to 0} u'(c) = \infty$ and $\lim_{c \to \infty} u'(c) = 0$ hold, ensuring interior solutions away from zero consumption; and (iii) constraint sets are convex with non-empty interior. These conditions ensure first-order conditions are necessary for optimality at interior solutions.

A key insight simplifies this apparently formidable problem: although the household makes all three decisions simultaneously from an economic perspective, we can organize the solution method so that decisions are solved sequentially, with each stage using information from the next. This is not merely a computational trick; it reflects the natural dependence structure of the problem. The labor-leisure choice determines market resources; given those resources, the consumption-saving choice determines liquid assets; given liquid assets, the portfolio choice follows. By respecting this structure, each stage becomes a tractable subproblem amenable to EGM.

We can make a few choices to create a sequential problem which will allow us to use multiple EGM steps in succession. First, the agent decides their labor-leisure trade-off and receives a wage. Their wage plus their previous bank balance then becomes their market resources. Second, given market resources, the agent makes a pure consumption-saving decision. Finally, given an amount of savings, the consumer then decides their risky portfolio share.

Starting from the beginning of the period, we can define the labor-leisure problem as

$$\begin{aligned} v_t^0(b_t, \theta_t) &= \max_{z_t} h(z_t) + v_t^1(m_t) \\ &\text{s.t.} \\ z_t &\in [0, 1] \\ \ell_t &= 1 - z_t \\ m_t &= b_t + \theta_t \ell_t. \end{aligned} \tag{5}$$

The pure consumption-saving problem is then

$$\begin{aligned} v_t^1(m_t) &= \max_{c_t} u(c_t) + \beta v_t^2(a_t) \\ &\text{s.t.} \\ c_t &\in [0, m_t] \\ a_t &= m_t - c_t. \end{aligned} \tag{6}$$

Finally, the risky portfolio problem is

$$\begin{aligned} v_t^2(a_t) &= \max_{\varsigma_t} \mathbb{E}_t \left[\Gamma_{t+1}^{1-\rho} v_{t+1}^0(b_{t+1}, \theta_{t+1}) \right] \\ &\text{s.t.} \\ \varsigma_t &\in [0, 1] \\ \mathbb{R}_{t+1} &= \mathsf{R} + (\mathbf{R}_{t+1} - \mathsf{R})\varsigma_t \\ b_{t+1} &= a_t \mathbb{R}_{t+1} / \Gamma_{t+1}. \end{aligned} \tag{7}$$

This sequential approach is explicitly modeled after the nested approaches explored in Clausen and Strub [2020] and Druedahl [2021]. However, we offer additional insights that expand on these methods. An important observation is that now, every single choice is self-contained in a subproblem, and although the structure is specifically chosen to minimize the number of state variables at every stage, the problem does not change by this structural imposition. This sequential formulation preserves the original problem because no uncertainty resolves between subproblems within a single period. From the agent's information set at time t, all three decisions are made simultaneously before any time-$t+1$ shocks realize. The expectation operator appears only in the final subproblem, ensuring decisions are made under identical information. From the perspective of the consumer, these decisions are essentially simultaneous, but a careful organization into sub-period problems enables us to solve the model more efficiently and can provide key economic insights. In this problem, as we will see, a key insight will be the ability to explicitly calculate the marginal value of wealth and the Frisch elasticity of labor.

2.2 Sequential Solution

As useful as it is to be able to use the EGM step more than once, there are clear problems where the EGM step is not applicable. This basic labor-portfolio choice problem demonstrates where we can use an additional EGM step, and where we cannot. First, we go over a subproblem where we cannot use the EGM step.

In reorganizing the labor-portfolio problem into subproblems, we assigned the utility of leisure to the leisure-labor subproblem and the utility of consumption to the consumption-savings subproblem. There are no more separable convex utility functions to assign to this problem, and even if we re-organized the problem in a way that moved one of the utility functions into this subproblem, they would not be useful in solving this subproblem via EGM as there is no direct relation between the risky share of portfolio and consumption or leisure. Therefore, the only way to solve this subproblem is through standard convex optimization and root-finding techniques (which admit no separable utility and thus cannot use EGM directly).

Restating the problem in compact form gives

$$v_t^2(a_t) = \max_{\varsigma_t} \mathbb{E}_t \left[\Gamma_{t+1}^{1-\rho} v_{t+1}^0 \left(a_t(\mathsf{R} + (\mathbf{R}_{t+1} - \mathsf{R})\varsigma_t)/\Gamma_{t+1}, \theta_{t+1} \right) \right]. \tag{8}$$

The first-order condition with respect to the risky portfolio share is then

$$\mathbb{E}_t \left[\Gamma_{t+1}^{-\rho} \frac{\partial v_{t+1}^0}{\partial b} \left(b_{t+1}, \theta_{t+1} \right) \left(\mathbf{R}_{t+1} - \mathsf{R} \right) \right] = 0. \tag{9}$$

Finding the optimal risky share requires numerical optimization and root-solving of the first-order condition. To close out the problem, we can calculate the envelope condition as

$$\frac{dv_t^2}{da}(a_t) = \mathbb{E}_t \left[\Gamma_{t+1}^{-\rho} \frac{\partial v_{t+1}^0}{\partial b} \left(b_{t+1}, \theta_{t+1} \right) \mathbb{R}_{t+1} \right]. \tag{10}$$

This completes the portfolio stage solution.[4]

The consumption-saving EGM follows Carroll [2006] but we cover it for exposition. We can begin the solution process by restating the consumption-savings subproblem in a more compact form, substituting the market resources constraint and ignoring the no-borrowing constraint for now. The problem is:

$$v_t^1(m_t) = \max_{c_t} u(c_t) + \beta v_t^2(m_t - c_t). \tag{11}$$

The first-order condition with respect to c_t yields the familiar Euler equation:

$$\frac{du}{dc}(c_t) = \beta \frac{dv_t^2}{da}(m_t - c_t) = \beta \frac{dv_t^2}{da}(a_t) \tag{12}$$

Inverting this equation is the (first) EGM step.[5]

$$\mathfrak{c}_t(a_t) = \left(\frac{du}{dc} \right)^{-1} \left(\beta \frac{dv_t^2}{da}(a_t) \right) \tag{13}$$

Given the utility function above, the marginal utility of consumption and its inverse are

[4]An alternative formulation avoids taking expectations more than once. We could define the portfolio choice subproblem as $v_t^2(a_t) = \max_{\varsigma_t} v_t^{1,\text{alt}}(a_t, \varsigma_t)$ where $v_t^{1,\text{alt}}(a_t, \varsigma_t) = \mathbb{E}_t[\Gamma_{t+1}^{1-\rho} v_{t+1}^0(b_{t+1}, \theta_{t+1})]$ with $\mathbb{R}_{t+1} = \mathsf{R} + (\mathbf{R}_{t+1} - \mathsf{R})\varsigma_t$ and $b_{t+1} = a_t \mathbb{R}_{t+1}/\Gamma_{t+1}$. Given the next period's solution, we calculate the marginal value functions as ${v_t^{1,\text{alt}}}^a(a_t, \varsigma_t) = \mathbb{E}_t[\Gamma_{t+1}^{-\rho} {v_{t+1}^0}^b(b_{t+1}, \theta_{t+1})\mathbb{R}_{t+1}]$ and ${v_t^{1,\text{alt}}}^{\varsigma}(a_t, \varsigma_t) = \mathbb{E}_t[\Gamma_{t+1}^{-\rho} {v_{t+1}^0}^b(b_{t+1}, \theta_{t+1}) a_t(\mathbf{R}_{t+1} - \mathsf{R})]$. Both can be computed in one expectation step. The optimal risky share then satisfies ${v_t^{1,\text{alt}}}^{\varsigma}(a_t, \varsigma_t^*) = 0$ with envelope condition ${v_t^2}^a(a_t) = {v_t^{1,\text{alt}}}^a(a_t, \varsigma_t^*)$.

[5]Invertibility of $\frac{du}{dc}$ requires strict monotonicity, which holds when $\frac{d^2u}{dc^2} < 0$. For CRRA utility with $\rho > 0$, we have $\frac{d^2u}{dc^2}(c) = -\rho c^{-\rho-1} < 0$, ensuring a one-to-one mapping between marginal utility and consumption levels.

$$\frac{du}{dc}(c) = c^{-\rho} \qquad \left(\frac{du}{dc}\right)^{-1}(x) = x^{-1/\rho}. \tag{14}$$

Carroll [2006] demonstrates that by using an exogenous grid of [a] points we can find the unique $\mathfrak{c}_t([\mathrm{a}])$ that optimizes the consumption-saving problem.[6] The strict concavity of u and v_t^2 (inherited from the value function) combined with the convex constraint set ensures the first-order condition is both necessary and sufficient[7] for a unique optimum. Further, using the market resources constraint, we can recover the exact amount of market resources that is consistent with this consumption-saving decision as

$$\mathfrak{m}_t([\mathrm{a}]) = \mathfrak{c}_t([\mathrm{a}]) + [\mathrm{a}]. \tag{15}$$

This $\mathfrak{m}_t([\mathrm{a}])$ is the "endogenous" grid that is consistent with the exogenous decision grid [a]. Now that we have a $(\mathfrak{m}_t([\mathrm{a}]), \mathfrak{c}_t([\mathrm{a}]))$ pair for each $a \in [\mathrm{a}]$, we can construct an interpolating consumption function for market resources points that are off-the-grid.

The envelope condition[8] will be useful in the next section, but for completeness we define it here.

$$\frac{dv_t^1}{dm}(m_t) = \beta \frac{dv_t^2}{da}(a_t) = \frac{du}{dc}(c_t) \tag{16}$$

The labor-leisure subproblem can be restated more compactly as:

$$v_t^0(b_t, \theta_t) = \max_{z_t} h(z_t) + v_t^1(b_t + \theta_t(1 - z_t)) \tag{17}$$

The first-order condition with respect to leisure is

$$\frac{dh}{dz}(z_t) = \frac{dv_t^1}{dm}(m_t)\theta_t \tag{18}$$

The marginal utility of leisure and its inverse are

$$\frac{dh}{dz}(z) = \nu^{1-\rho} z^{-\zeta} \qquad \left(\frac{dh}{dz}\right)^{-1}(x) = (x/\nu^{1-\rho})^{-1/\zeta} \tag{19}$$

Using an exogenous grid of [m] and $[\theta]$, we can find leisure as

$$\mathfrak{z}_t([\mathrm{m}], [\theta]) = \left(\frac{dh}{dz}\right)^{-1}\left(\frac{dv_t^1}{dm}([\mathrm{m}])[\theta]\right) \tag{20}$$

However, agents with low market resources m_t and high wage offers θ_t may find the unconstrained optimum violates the feasibility constraint $z_t \in [0, 1]$. When this occurs, we project the solution onto the constraint boundary, defining the constrained optimal function $\hat{\mathfrak{z}}_t([\mathrm{m}], [\theta])$ as

$$\hat{\mathfrak{z}}_t([\mathrm{m}], [\theta]) = \max\left\{\min\left\{\mathfrak{z}_t([\mathrm{m}], [\theta]), 1\right\}, 0\right\} \tag{21}$$

[6] We adopt the notational convention that bracketed variables (e.g., [a], [m]) denote exogenous grids of points on which we evaluate expectations and marginal values, while gothic (fraktur) letters (e.g., $\mathfrak{c}$, $\mathfrak{m}$) denote endogenous quantities constructed by inverting first-order conditions. This visual distinction emphasizes that grids are chosen inputs while gothic variables emerge from the EGM inversion step.

[7] Necessity follows from standard optimality conditions under differentiability. Sufficiency follows from the strict concavity of the objective function, which guarantees that any critical point is a global maximum. The strict concavity of CRRA utility and the inheritance of concavity through the continuation value ensure uniqueness of the solution.

[8] Follows from the envelope theorem, valid when the value function is differentiable and the constraint set satisfies standard regularity conditions.

This projection ensures feasibility.[9] In regions where constraints bind, the Kuhn-Tucker conditions replace the unconstrained first-order condition. Care must be taken during interpolation to handle potential non-differentiabilities at constraint boundaries, though these typically affect only small regions of the state space.

Then, we derive labor as $\mathfrak{l}_t(m_t, \theta_t) = 1 - \hat{\mathfrak{z}}_t(m_t, \theta_t)$. Finally, for each θ_t and m_t as an exogenous grid, we can find the endogenous grid of bank balances as $\mathfrak{b}_t(m_t, \theta_t) = m_t - \theta_t \mathfrak{l}_t(m_t, \theta_t)$.

The envelope condition then provides the marginal value of bank balances as

$$v_t^{0^b}(b_t, \theta_t) = v_t^{1'}(m_t) = h'(z_t)/\theta_t. \tag{22}$$

This envelope condition, together with the first-order condition, implicitly defines the heterogeneous Frisch elasticity of labor supply, which varies across states (b_t, θ_t).[10]

The resulting endogenous grid for the labor-leisure problem is curvilinear rather than rectilinear, requiring specialized interpolation methods. We defer the detailed discussion of interpolation on curvilinear grids to Section ??.[11]

Having demonstrated how sequential decomposition works in a three-choice problem with one-dimensional state spaces, we now tackle the more challenging case where the state space itself is multidimensional.

3 The EGM^n in Higher Dimensions

The labor-portfolio problem demonstrates the power of sequential decomposition, but one might wonder whether the approach scales to genuinely high-dimensional state spaces. After all, many economically important problems (retirement planning with multiple accounts, durable goods choices, human capital investment) involve multiple state variables that must be tracked simultaneously. The pension deposit problem we now examine shows that EGM^n can handle such complexity, though it requires confronting a new challenge: how to interpolate on grids that lose even their topological regularity.

The problem in Section ?? demonstrates the simplicity of solving problems sequentially. However, as constructed, the problem has only one state variable and one post-decision state variable per stage. Can EGM^n be used to solve higher dimensional problems? In short, yes, but it requires additional thought on interpolation.

3.1 A more complex problem

For a demonstration, we now turn to the problem of a worker saving up for retirement. This worker must consume, save, and deposit resources into a tax-advantaged account that can not be liquidated until retirement. In the recursive problem, the worker begins a new period with a liquid account of market resources m_t and an illiquid account of retirement savings n_t. The worker maximizes their utility by choosing consumption c_t and pension deposit d_t. The pension deposit is set aside on a retirement account that is exposed to a risky return, while their post-consumption liquid assets accrue risk-free interest every period. The worker additionally receives an income that faces a permanent (Γ_{t+1}) and a transitory (θ_{t+1}) shock every period. At the age of 65, the worker is retired and their assets are liquidated, at which point the

[9] At the lower bound $z_t = 0$, the Kuhn-Tucker condition is $h'(0) \leq v_t^{1'}(m_t)\theta_t$, with complementary slackness ensuring the constraint binds only when the marginal utility of leisure is insufficient to justify reduced labor supply. Similarly, at $z_t = 1$, the agent chooses full leisure only when $h'(1) \geq v_t^{1'}(m_t)\theta_t$.

[10] The Frisch elasticity of labor supply is defined as $\varepsilon_{\ell,\theta} = \frac{\partial \ell}{\partial \theta}\frac{\theta}{\ell}$ holding the marginal utility of wealth constant. From the first-order condition $h'(z_t) = v_t^{1'}(m_t)\theta_t$, we implicitly differentiate with respect to θ_t while holding $v_t^{1'}(m_t)$ fixed: $h''(z_t)\frac{\partial z_t}{\partial \theta} = v_t^{1'}(m_t)$. Since $\ell_t = 1 - z_t$, we obtain $\frac{\partial \ell_t}{\partial \theta} = -\frac{v_t^{1'}(m_t)}{h''(1-\ell_t)}$. For the CRRA leisure utility, $h''(z) = -\zeta\nu^{1-\rho}z^{-\zeta-1} < 0$, making the derivative positive. The elasticity $\varepsilon_{\ell,\theta} = -\frac{v_t^{1'}(m_t)}{h''(1-\ell_t)}\frac{\theta_t}{\ell_t}$ varies with the state because both $h''(1-\ell_t)$ and the ratio θ_t/ℓ_t depend on (b_t, θ_t).

[11] The labor-leisure problem could be solved using simpler interpolation methods since the grid warping occurs along only one dimension (wage offers). However, we use Warped Grid Interpolation here for two pedagogical reasons: (1) it demonstrates the sequential decomposition that is the essence of EGM^n, and (2) it illustrates WGI in a transparent setting. WGI is robust to various types of grid warping—from simple one-dimensional stretching to complex multidimensional distortions—making it valuable to understand in this simpler context before encountering the genuinely unstructured grids of Section ??.

state reduces to one liquid account of market resources. The feasibility constraint $m_t \geq c_t + d_t$ ensures non-negative liquid savings. The worker's recursive problem is:

$$\begin{aligned} v_t^0(m_t, n_t) &= \max_{c_t, d_t} u(c_t) + \beta \mathbb{E}_t \left[\Gamma_{t+1}^{1-\rho} v_{t+1}^0(m_{t+1}, n_{t+1}) \right] \\ \text{s.t.} &\quad c_t \geq 0, \quad d_t \geq 0 \\ a_t &= m_t - c_t - d_t \\ b_t &= n_t + d_t + \mathrm{g}(d_t) \\ m_{t+1} &= a_t \mathsf{R} / \Gamma_{t+1} + \theta_{t+1} \\ n_{t+1} &= b_t \mathbf{R}_{t+1} / \Gamma_{t+1} \end{aligned} \tag{23}$$

where

$$\mathrm{g}(d) = \chi \log(1 + d). \tag{24}$$

This problem can subsequently be broken down into 3 stages: a pension deposit stage, a consumption stage, and an income shock stage.

3.2 Sequential Decomposition

In the deposit stage, the worker begins with market resources and a retirement savings account. The worker must maximize their value of liquid wealth l_t and retirement balance b_t by choosing a pension deposit d_t, which must be positive. The retirement balance b is the cash value of their retirement account plus their pension deposit and an additional amount $\mathrm{g}(d_t)$ that provides an incentive to save for retirement. As we'll see, this additional term will allow us to use the Endogenous Grid Method to solve this subproblem.

$$\begin{aligned} v_t^0(m_t, n_t) &= \max_{d_t} v_t^1(l_t, b_t) \\ \text{s.t.} &\quad d_t \geq 0 \\ l_t &= m_t - d_t \\ b_t &= n_t + d_t + \mathrm{g}(d_t) \end{aligned} \tag{25}$$

After making their pension decision, the worker begins their consumption stage with liquid wealth l_t and retirement balance b_t. From their liquid wealth, the worker must choose a level of consumption to maximize utility and continuation value w_t. After consumption, the worker is left with post-decision states that represent liquid assets a_t and retirement balance b_t, which passes through this problem unaffected because it can't be liquidated until retirement.

$$\begin{aligned} v_t^1(l_t, b_t) &= \max_{c_t} u(c_t) + \beta \mathrm{w}_t(a_t, b_t) \\ \text{s.t.} &\quad c_t \geq 0 \\ a_t &= l_t - c_t \end{aligned} \tag{26}$$

Finally, the post-decision value function w_t represents the value of both liquid and illiquid account balances before the realization of uncertainty regarding the risky return and income shocks. Since we are dealing with a normalized problem, this stage handles the normalization of state variables and value functions into the next period.

$$\begin{aligned} \mathrm{w}_t(a_t, b_t) &= \mathbb{E}_t \left[\Gamma_{t+1}^{1-\rho} v_{t+1}^0(m_{t+1}, n_{t+1}) \right] \\ \text{s.t.} &\quad a_t \geq 0, \quad b_t \geq 0 \\ m_{t+1} &= a_t \mathsf{R} / \Gamma_{t+1} + \theta_{t+1} \\ n_{t+1} &= b_t \mathbf{R}_{t+1} / \Gamma_{t+1} \end{aligned} \tag{27}$$

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The advantage of conceptualizing this subproblem as a separate stage is that we can construct a function w_t and use it in the prior optimization problems without having to worry about stochastic optimization and taking expectations repeatedly.

3.3 Solution via Sequential EGM

As seen in the consumption stage above, the retirement balance b_t passes through the problem unaffected because it can't be liquidated until retirement. In this sense, it is already a post-decision state variable. To solve this problem, we can use a fixed grid of [b] and for each obtain endogenous consumption and ex-ante market resources using the simple Endogenous Grid Method for the consumption problem.

In the deposit stage, both the state variables and post-decision variables are different since both are affected by the pension deposit decision.

First, we can rewrite the pension deposit problem more compactly:

$$v_t^0(m_t, n_t) = \max_{d_t} v_t^1(m_t - d_t, n_t + d_t + \mathrm{g}(d_t)) \tag{30}$$

The first-order condition is

$$\frac{\partial v_t^1}{\partial l}(l_t, b_t)(-1) + \frac{\partial v_t^1}{\partial b}(l_t, b_t)(1 + \frac{d\mathrm{g}}{dd}(d_t)) = 0. \tag{31}$$

This condition is necessary for interior optima, with sufficiency following from the concavity of v_t^1 inherited from the continuation value function and the convexity of the constraint set.[12] Rearranging yields

$$\frac{d\mathrm{g}}{dd}(d_t) = \frac{\frac{\partial v_t^1}{\partial l}(l_t, b_t)}{\frac{\partial v_t^1}{\partial b}(l_t, b_t)} - 1 \tag{32}$$

where

[12] The concavity of v_t^1 in both arguments ensures the objective function is concave in d_t, making any critical point a global maximum. The constraint $d_t \geq 0$ is handled by checking feasibility of the unconstrained solution.

$$\frac{d\mathrm{g}}{dd}(d) = \frac{\chi}{1+d} \qquad \left(\frac{d\mathrm{g}}{dd}\right)^{-1}(y) = \chi/y - 1 \tag{33}$$

Note that $\frac{d\mathrm{g}}{dd}(d) > 0$ for all $d > -1$, ensuring strict monotonicity and hence invertibility. We can find

$$\mathfrak{d}_t(l_t, b_t) = \left(\frac{d\mathrm{g}}{dd}\right)^{-1}\left(\frac{\frac{\partial v_t^1}{\partial l}(l_t, b_t)}{\frac{\partial v_t^1}{\partial b}(l_t, b_t)} - 1\right) \tag{34}$$

Using this, we can back out n_t as

$$\mathfrak{n}_t(l_t, b_t) = b_t - \mathfrak{d}_t(l_t, b_t) - \mathrm{g}(\mathfrak{d}_t(l_t, b_t)) \tag{35}$$

and m_t as

$$\mathfrak{m}_t(l_t, b_t) = l_t + \mathfrak{d}_t(l_t, b_t) \tag{36}$$

In sum, given an exogenous grid ([l], [b]) we obtain the triple $(\mathfrak{m}_t(l_t, b_t), \mathfrak{n}_t(l_t, b_t), \mathfrak{d}_t(l_t, b_t))$, which we can use to create an interpolator for the decision rule d_t.

To close the solution method, the envelope conditions are

$$\begin{aligned} \frac{\partial v_t^0}{\partial m}(m_t, n_t) &= \frac{\partial v_t^1}{\partial l}(l_t, b_t) \\ \frac{\partial v_t^0}{\partial n}(m_t, n_t) &= \frac{\partial v_t^1}{\partial b}(l_t, b_t). \end{aligned} \tag{37}$$

The resulting endogenous grid in this problem is irregular and unstructured, unlike the curvilinear grid in the labor-leisure problem of Section ??. The interpolation techniques required for this more complex case are discussed in detail in Section ??.

The pension deposit problem reveals a new challenge: even when we successfully apply EGM at each stage, the resulting endogenous grids may be highly irregular. This brings us to the interpolation problem.

4 Multivariate Interpolation on Non-Rectilinear Grids

EGM's efficiency comes from working on exogenous grids where calculations are straightforward, then using the inverted Euler equation to recover endogenous state variables. This creates an interpolation challenge: the resulting endogenous grid is warped, stretched, or even unstructured. Standard multilinear interpolation assumes a rectilinear grid where each dimension varies independently, so it simply fails on EGM-generated grids. We need interpolation methods that respect the actual geometry of these grids. For curvilinear grids that preserve topological structure, Warped Grid Interpolation exploits that structure for efficiency. For fully unstructured grids where even topology breaks down, we turn to Gaussian Process Regression.

This section presents alternative interpolation methods for non-rectilinear grids. We distinguish between two cases that arise in EGM applications: curvilinear grids that retain regular topological structure, and fully unstructured grids where this regularity is lost. For the former, we present Warped Grid Interpolation (WGI), which improves upon the interpolation in White [2015]. For the latter, we present a machine learning approach based on Gaussian Process Regression as in Scheidegger and Bilionis [2019].

4.1 Motivation for Alternative Methods

Unstructured interpolation arises in many dynamic programming applications when using the Endogenous Grid Method because the first-order conditions might be highly non-linear and non-monotonic, or because boundary constraints induce kinks in the policy and value functions. In the pure consumption-savings problem, a one-dimensional exogenous grid of post-decision liquid assets [a] results in a one-dimensional endogenous grid of total market resources [m]. However, as we know from standard EGM, the spacing in the [m] grid differs from the spacing in the [a] grid as the inverted Euler equation is non-linear. This is no problem in a one-dimensional problem as we can simply use non-uniform linear interpolation.

However, the same is true of higher dimensional problems, where the exogenous grid gets mapped to a warped endogenous grid. In these cases, the grid points generated by the EGM step are not evenly spaced, leading to the need for specialized interpolation. When the endogenous grid retains its topological structure despite the warping, we can use efficient curvilinear interpolation methods. When this structure is lost entirely, as in the pension deposit problem of Section ??, we must resort to unstructured interpolation methods.

A similar approach using Delaunay triangulation was presented in Ludwig and Schön [2018]. However, this approach is not well suited for our purposes because triangulation can be computationally intensive and slow (often offsetting the efficiency gains from the Endogenous Grid Method). As an alternative, we introduce the use of Gaussian Process Regression (GPR) along with the Endogenous Grid Method for unstructured grids. GPR is computationally efficient, and tools exist to easily parallelize and take advantage of hardware such as Graphics Processing Units (GPU) [Gardner et al., 2018].

4.2 Interpolation on Curvilinear Grids

We begin with the case of curvilinear grids, which arise in problems like the labor-leisure example in Section ??. In this case, standard multi-linear interpolation is inapplicable because the resulting endogenous grid is non-rectilinear. Instead, we introduce Warped Grid Interpolation (WGI), which exploits the preserved topological structure of curvilinear grids to achieve computational efficiency superior to triangulation-based or dense interpolation methods.

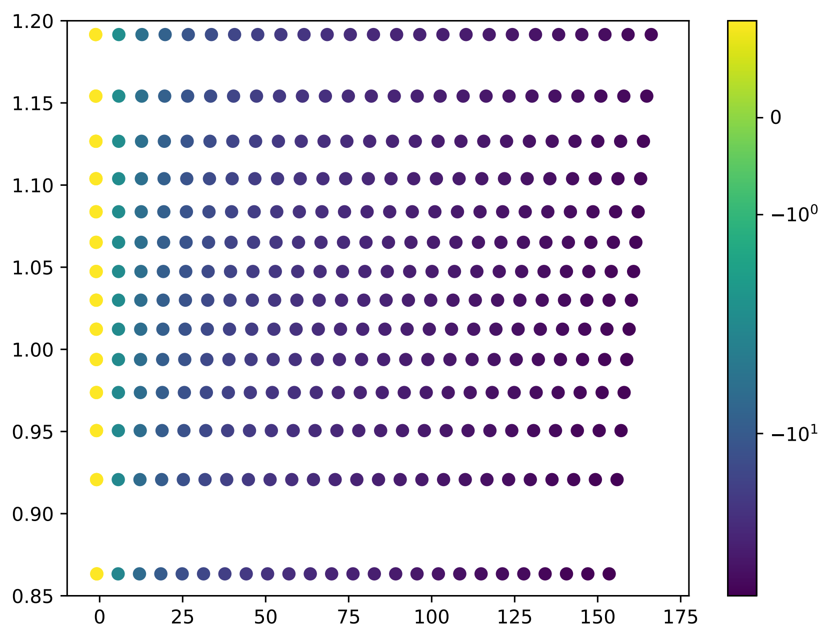

Figure 1: Warped curvilinear grid that results from multivariate EGM. This grid can be interpolated by WGI.

Definition 4.1 (Curvilinear Grid)**.** *A grid of points* $\{(x_{ij}, y_{ij}) : i = 1, \ldots, I, \; j = 1, \ldots, J\}$ *in* $\mathbb{R}^2$ *is curvilinear if there exists a continuous and piecewise differentiable mapping* $\phi : [0, I-1] \times [0, J-1] \to \mathbb{R}^2$ *such that* $\phi(i, j) = (x_{ij}, y_{ij})$ *and* ϕ *is locally invertible almost everywhere.[ĉurvilinear-def] The grid is topologically regular if* ϕ *preserves the ordering of indices: for adjacent indices* (i, j) *and* (i', j') *in the index space, the corresponding points in physical space remain geometrically adjacent.*

Consider a function $f : \mathbb{R}^2 \to \mathbb{R}$ for which we observe values $z_{ij} = f(x_{ij}, y_{ij})$ at a curvilinear grid of points $\{(x_{ij}, y_{ij})\}_{i,j}$. The points (x_{ij}, y_{ij}) are not evenly spaced and do not form a rectilinear grid, yet they retain their matrix structure through the continuous mapping ϕ from index space (i, j) to physical space (x, y). This topological regularity is crucial: neighborhood relationships are preserved even though Euclidean distances are distorted.

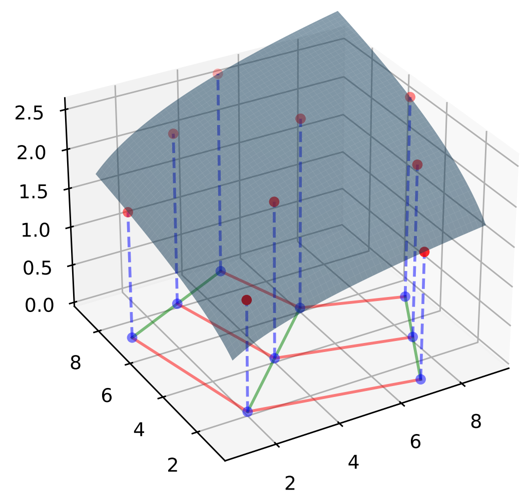

Figure 2: True function and curvilinear grid of points for which we know the value of the function.

Figure 2 displays the true function in three-dimensional space along with the observed grid points. Connecting points along each row and column reveals the piecewise affine structure characteristic of curvilinear grids. The key mathematical insight is that there exists a homotopy—a continuous deformation—between the curvilinear grid in physical space and the rectilinear grid in index space, as shown in Figure 3.

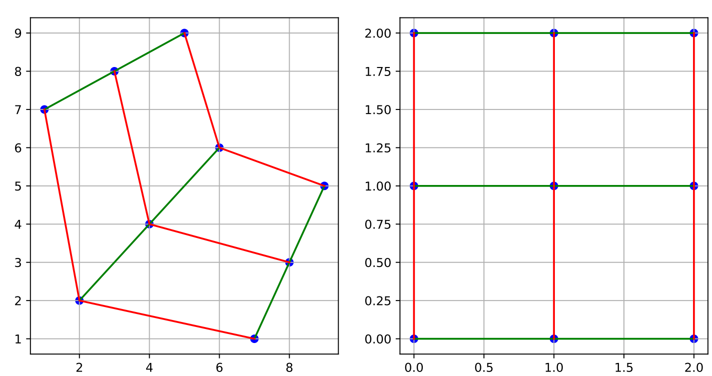

Figure 3: Homotopy between the curvilinear grid and the index coordinates of the matrix.

The choice of dimension order (rows versus columns) can affect accuracy when the mapping ϕ is highly anisotropic. Figure 4 illustrates this process: the vertical line at $x = x^*$ intersects each column at specific y-coordinates (shown as circles), and interpolating across these intersection points yields the final estimate. The method leverages the grid's topological structure by using fast one-dimensional interpolation sequentially, avoiding the computational cost of triangulation.[ŵgi-examples]

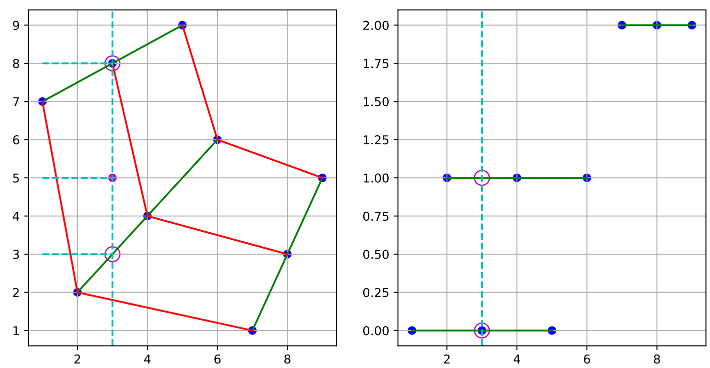

Figure 4: WGI intersects the horizontal line at $y = y^*$ with each row of the curvilinear grid (circles), then interpolates across these intersections to evaluate $f(x^*, y^*)$.

WGI exploits the preserved matrix structure of curvilinear grids to achieve $O(I+J)$ complexity per evaluation, compared to $O(IJ\log(IJ))$ for Delaunay triangulation construction or $O((IJ)^2)$ for dense grid methods.[13] Moreover, WGI naturally extends to higher dimensions through recursive application, whereas triangulation-based methods suffer from the curse of dimensionality in simplex construction.

4.3 Interpolation on Unstructured Grids

We now turn to the more challenging case of fully unstructured grids, as arise in the pension deposit problem of Section ??. In this case, the endogenous grid loses even its topological regularity, making curvilinear interpolation methods inapplicable. We use Gaussian Process Regression to handle this case.

Definition 4.2 (Unstructured Grid)**.** *A grid of points $\{(\mathbf{x}_k, z_k) : k = 1, \ldots, N\}$ in $\mathbb{R}^d \times \mathbb{R}$ is unstructured if it cannot be represented by a continuous mapping from a regular index space that preserves local neighborhoods. Formally, no continuous and piecewise differentiable mapping $\phi : [0, n_1 - 1] \times \cdots \times [0, n_d - 1] \to \mathbb{R}^d$ exists such that the grid points correspond to ϕ evaluated at integer lattice points while preserving adjacency relationships.[ûnstructured-def] Equivalently, the grid is unstructured when points that are neighbors in physical space may correspond to arbitrarily distant indices in any attempted regular indexing scheme, or when multiple grid points map to overlapping regions of the domain.*

Starting from a regular and rectilinear exogenous grid of liquid assets post-consumption l_t and pension balances post-deposit b_t shown in Figure 5, we obtain Figure 6 which shows an irregular and unstructured endogenous grid of market resources m_t and pension balances pre-deposit n_t.

On unstructured grids, multiple exogenous points may map to overlapping regions of the endogenous space.[14] To interpolate a function defined on such grids, we use Gaussian Process Regression (GPR) as in Scheidegger and Bilionis [2019]. GPR handles this by finding the maximum likelihood function given all observed points, effectively averaging over conflicting information in a principled manner.

A Gaussian Process (GP) is a collection of random variables, any finite subset of which has a joint Gaussian distribution. Formally, a GP is completely specified by its mean function $m(\mathbf{x})$ and covariance (kernel) function $k(\mathbf{x}, \mathbf{x}')$. For any finite set of points, we have

$$\begin{aligned} \mathbf{X} \sim \mathcal{N}(\mu, \mathbf{\Sigma}) \quad \text{s.t.} \quad x_i \sim \mathcal{N}(\mu_i, \sigma_{ii}) \\ \text{and} \quad \sigma_{ij} = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)] \quad \forall i, j \in \{1, \ldots, n\}. \end{aligned} \tag{38}$$

[13]The complexity analysis assumes binary search for locating the relevant grid cell in each dimension. In practice, if evaluations are performed along a continuous path (as in policy function iteration), the search can be accelerated using the previous evaluation's location as an initial guess, reducing amortized complexity.

[14]Unlike Delaunay triangulation, which requires preprocessing to remove duplicate or nearly-collinear points, GPR naturally accommodates such degeneracies through its probabilistic framework. The posterior mean provides a smooth approximation even when the endogenous grid contains local irregularities or overlapping regions.

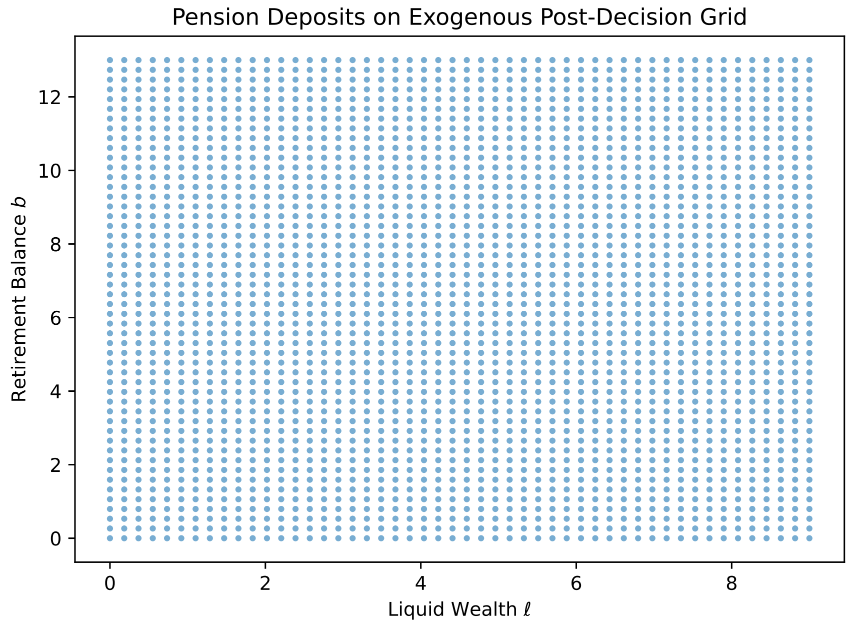

Figure 5: A regular, rectilinear exogenous grid of pension balances after deposit b_t and liquid assets after consumption l_t.

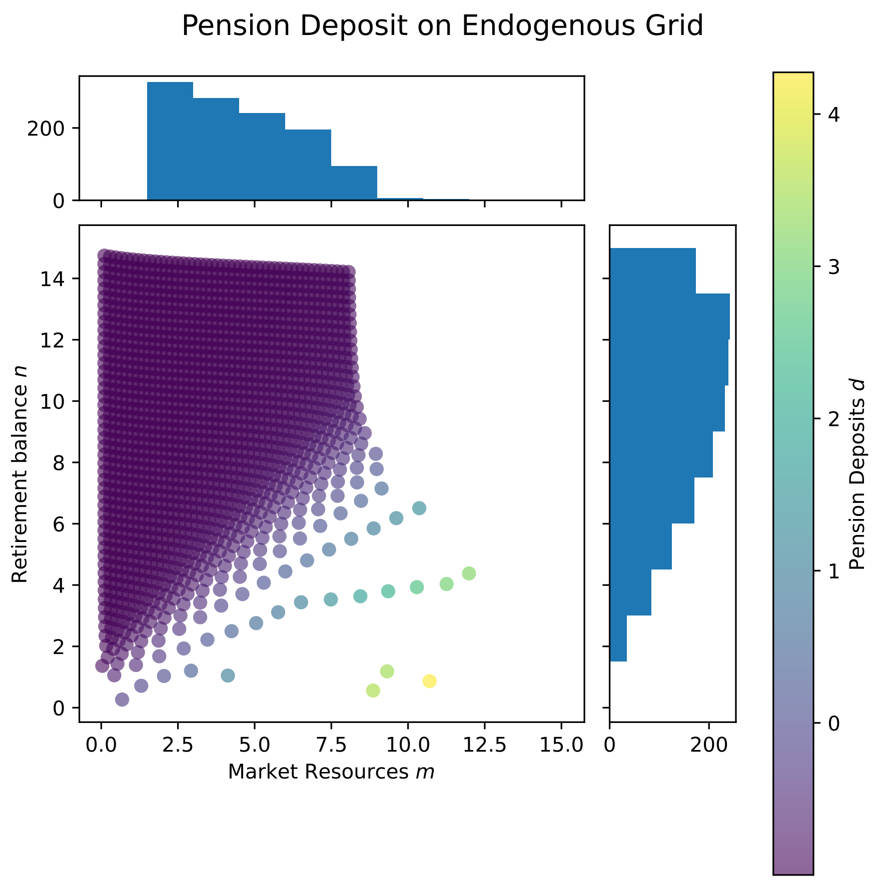

Figure 6: An irregular, unstructured endogenous grid of market resources m_t and pension balances before deposit n_t.

where

$$\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix} \quad \mathbf{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_{nn} \end{bmatrix}. \tag{39}$$

A Gaussian Process can be used to represent a probability distribution over the space of functions in n dimensions. Thus, Gaussian Process Regression (GPR) finds the posterior distribution over functions given observed data points. The posterior distribution is

$$\mathbb{P}(\mathbf{f}|\mathbf{X}) = \mathcal{N}(\mathbf{f}|\mathbf{m}, \mathbf{K}) \tag{40}$$

where $\mathbf{f}$ is the vector of function values at the points $\mathbf{X}$, $\mathbf{m}$ is the posterior mean vector, and $\mathbf{K}$ is the covariance matrix determined by the kernel function $k(\cdot, \cdot)$ that describes the covariance between function values at different points.

A standard kernel function is the squared exponential (or radial basis function) kernel, which is defined as

$$k(\mathbf{x}_i, \mathbf{x}_j) = \sigma_f^2 \exp\left(-\frac{1}{2l^2}(\mathbf{x}_i - \mathbf{x}_j)'(\mathbf{x}_i - \mathbf{x}_j)\right) \tag{41}$$

where σ_f^2 is the signal variance and l is the length-scale parameter. This kernel is infinitely differentiable and assumes smooth functions. Using GPR to interpolate a function f, we can both predict the value of the function at a point $\mathbf{x}_*$ and quantify the uncertainty in the prediction via the posterior variance, which provides useful information about approximation accuracy.

In Figure 7, we see the function we are trying to approximate along with a sample of data points for which we know the value of the function. In practice, the value of the function is unknown and/or expensive to compute, so we must use a limited amount of data to approximate it.

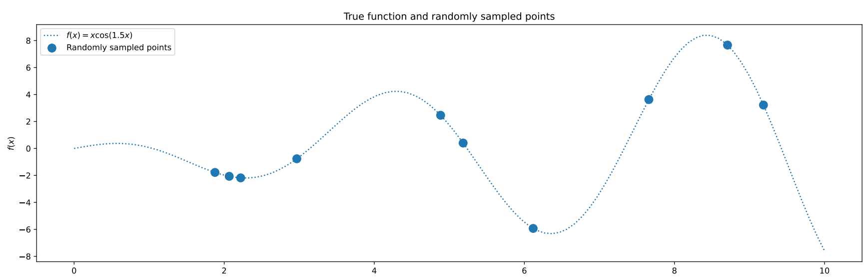

Figure 7: The true function that we are trying to approximate and a sample of data points.

A Gaussian Process is an infinite dimensional random process which can be used to represent a probability distribution over the space of functions. In Figure 8, we see a random sample of functions from the GPR posterior, which is a Gaussian Process conditioned on fitting the data. From this small sample of functions, we can see that the GP generates functions that fit the data well, and the goal of GPR is to find the one function that best fits the data given some hyperparameters by minimizing the negative log-likelihood of the data.

In Figure 9, we see the result of GPR with a particular parametrization[15] of the kernel function. The dotted line shows the true function, while the blue dots show the known data points. GPR provides the mean function which best fits the data, represented in the figure as an orange line. The shaded region represents a 95% confidence interval, which is the uncertainty of the predicted function. Along with finding the best

[15]The specific hyperparameters (signal variance σ_f^2 and length-scale l) are optimized by maximizing the marginal likelihood of the observed data. Implementation details and code for reproducing these figures are provided in the accompanying computational notebooks.

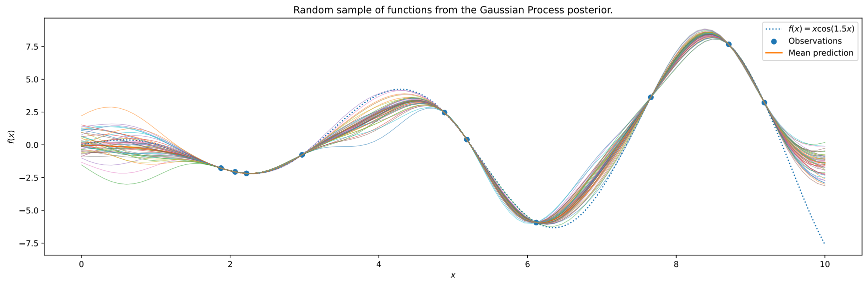

Figure 8: A random sample of functions from the GPR posterior that fit the data. The goal of GPR is to find the function that best fits the data.

fit of the function, GPR provides the uncertainty of the prediction, which is useful information as to the accuracy of the approximation.

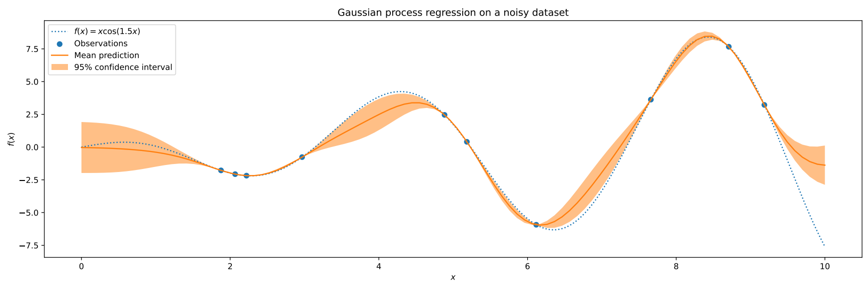

Figure 9: GPR finds the function that best fits the data given some hyperparameters. GPR then optimizes over the parameter space to find the function that minimizes the negative log-likelihood of the data.

5 Conditions for using the Sequential Endogenous Grid Method

A researcher confronting a new multidecision problem faces a practical question: does my problem admit sequential EGM, and if so, how should I decompose it? This section provides operational guidance. We identify the mathematical structures that make EGM applicable (separability in utility, invertibility in transitions), and equally importantly, we explain how to order subproblems to minimize state variable proliferation. Poor sequencing can inadvertently complicate the problem; good sequencing exploits the natural information structure.

5.1 Splitting the problem into subproblems

The first step in using the Sequential Endogenous Grid Method is to split the problem into subproblems. This process of splitting up the problem has to be strategic to not insert additional complexity into the original problem. If one is not careful when doing this, the subproblems can become more complex and intractable than the original problem.

To split up the problem, we first count the number of control variables or decisions faced by the agent. Ideally, if the agent has n control variables, then the problem should be split into n subproblems, each handling a different control variable. For counting the number of control variables, it is important to not double count variables which are equivalent and have market clearing conditions. For example, the decision of how much to consume and how much to save may seem like two different choices, but because of the market clearing condition $c + a = m$ they are resolved simultaneously and count as only one decision variable. Similarly, the choice between labor and leisure are simultaneous and count as only one decision.

Having counted our control variables, we look for differentiable and invertible utility functions which are separable in the dynamic programming problem, such as in Section ?? of the paper, or differentiable and invertible functions in the transition, as in Section ?? of the paper. In Section ??, we have additively separable utility of consumption and leisure, which allows for each of these control variables to be handled by separate subproblems. So, it makes sense to split the utility between subproblems and attach one to the consumption subproblem and one to the leisure subproblem.

As mentioned in that section, however, there are only two separable utility functions in the problem which have been assigned to two subproblems already. This leaves one control variable without a separable utility function. In that case, there is not another Endogenous Grid Method step to exploit, and this subproblem has to be handled by standard convex optimization techniques such as maximization of the value function (VFI) or finding the root of the Euler equation (PFI).

Now that we have split the problem into conceptual subproblems, it is important to sequence them in such a way that they don't become more complex than the original problem. The key here is to avoid adding unnecessary state variables. For example, in the consumption-leisure-portfolio problem, if we were to choose consumption first, we would have to track the wage rate into the following leisure subproblem. This would mean that our consumption problem would be two-dimensional as well as our labor decision problem. As presented, the choice of order in Section ?? ensures that the consumption problem is one-dimensional, as we can shed the information about the wage rate offer after the agent has made their labor-leisure decision. If we did this the other way, the problem would be more complex and require additional computational resources.[16]

Therefore, strategic ordering of subproblems can greatly simplify the solution process and reduce the computational burden.

Consider the utility function of the form

$$\mathrm{U}(a) = \mathrm{u}_{-i}(a^{-i}) + \mathrm{u}_i(a^i) \tag{42}$$

where a^i is the i-th control variable and a^{-i} is the vector of all control variables except the i-th one. This utility function is separable in the control variables that correspond to the index i.

$$\begin{aligned} \mathrm{V}(x, s) = \max_{a \in \Gamma(x,s)} & \mathrm{U}(a) + \beta \mathbb{E}\left[\mathrm{V}'(x', s') | y, s\right] \\ \text{s.t.} & \\ y &= \mathrm{T}(x, a) \\ x' &= \mathrm{G}(y, s) \end{aligned} \tag{43}$$

For simplicity, define

$$\mathrm{W}(y, s) = \beta \mathbb{E}\left[\mathrm{V}'(\mathrm{G}(y, s), s') | y, s\right] \tag{44}$$

then

$$\begin{aligned} \mathrm{V}(x, s) = \max_{a \in \Gamma(x,s)} & \mathrm{U}(a) + \mathrm{W}(y, s) \\ \text{s.t.} & \\ y &= \mathrm{T}(x, a) \end{aligned} \tag{45}$$

the first-order condition (assuming an interior solution)

$$\frac{\partial \mathrm{U}(a)}{\partial a^i} + \sum_{j=1}^{n} \frac{\partial \mathrm{W}(y, s)}{\partial y^j} \frac{\partial \mathrm{T}^j(x, a)}{\partial a^i} = 0 \tag{46}$$

we require $\frac{\partial \mathrm{T}^j(x,a)}{\partial a^i} = 0$ for $j \neq i$ (separability in the transition) to be able to solve for a^i independently.

[16] To see this concretely, the consumption subproblem would become two-dimensional: $v^0(b, \theta) = \max_c \mathrm{u}(c) + v^1(b', \theta)$ subject to $b' = b - c \geq -\theta$, requiring interpolation on a (b, θ) grid instead of just b. The labor-leisure subproblem would then have the additional constraint: $v^1(b', \theta) = \max_z h(z) + v^2(a)$ subject to $0 \leq z \leq 1$ and $a = b' + \theta(1 - z) \geq 0$. The poor ordering forces us to carry the wage state through both stages, doubling the dimensionality of the first stage.

$$\frac{\partial \mathrm{U}(a)}{\partial a^i} + \frac{\partial \mathrm{W}(y,s)}{\partial y^i}\frac{\partial \mathrm{T}^i(x,a)}{\partial a^i} = 0 \tag{47}$$

In Section ??, we see that a problem with a differentiable and invertible transition can also be used to embed an additional Endogenous Grid Method step. Because the transition applies independently to a state variable that is not related to the other control variable, consumption, it can be handled separately from the consumption subproblem. In this particular problem, however, it turns out to make no difference how we order the two subproblems. This is because the control variables, consumption and pension deposit, each affect a separate resource account, namely market resources and pension balance. Because of this, the two subproblems are independent of each other and can be solved in any order.

A good rule of thumb is that when splitting up a problem into subproblems, we should try to reduce the information set that is passed onto the next subproblem. In Section ??, choosing leisure-labor and realizing total market resources before consumption allows us to shed the wage rate offer state variable before the consumption problem, and we know that for the portfolio choice we only need to know liquid assets after expenditures (consumption). Thus, the order makes intuitive sense; agent first chooses leisure-labor, realizing total market resources, then chooses consumption and savings, and finally chooses their risky portfolio choice. In Section ??, there are two expenditures that are independent of each other, consumption and deposit, and making one decision or the other first does not reduce the information set for the agent, thus the order of these subproblems does not matter.

5.2 The Endogenous Grid Method for Subproblems

Once we have strategically split the problem into subproblems, we can use the Endogenous Grid Method in each applicable subproblem while iterating backwards from the terminal period. As demonstrated in Section ?? and Section ??, the EGM step can be applied when there is a separable, differentiable and invertible utility function in the subproblem or when there is a differentiable and invertible transition in the subproblem. We will discuss each of these cases in turn.

Proposition 5.1 (Applicability with Separable Utility)**.** ***EGM***
A generic subproblem with a differentiable and invertible utility function can be characterized as follows:

$$\begin{aligned} \mathrm{V}(x) &= \max_{a\in\Gamma(x)} \mathrm{U}(x,a) + \mathrm{W}(y) \\ &s.t. \\ y &= \mathrm{T}(x,a) \end{aligned} \tag{48}$$

where $\mathrm{W}(y) = \beta\mathbb{E}[\mathrm{V}'(y)]$ *is the continuation value. For an interior solution, the first-order condition is*

$$\frac{\partial \mathrm{U}(x,a)}{\partial a} + \frac{d\mathrm{W}}{dy}(y)\frac{\partial \mathrm{T}(x,a)}{\partial a} = 0 \tag{49}$$

When corner solutions occur (e.g., a at constraint boundaries), the unconstrained optimum from inverting the first-order condition must be projected onto the feasible set, as demonstrated in Section ?? for the leisure choice. For interior solutions where the marginal utility is strictly monotone,[êgm-invertibility] we can invert to obtain the Endogenous Grid Method step:

$$a = \left(\frac{\partial \mathrm{U}(x,a)}{\partial a}\right)^{-1}\left[-\frac{d\mathrm{W}}{dy}(y)\frac{\partial \mathrm{T}(x,a)}{\partial a}\right] \tag{50}$$

By using an exogenous grid of the post-decision state y, we can solve for the optimal decision rule a at each point on the grid. This is the Endogenous Grid Method step. Uniqueness of the solution is ensured when the utility function is strictly concave in a.

5.3 Applicability to Transition Functions

Proposition 5.2 (Applicability with Invertible Transitions)**.** ***EGM***
If the generic subproblem has no separable utility, but instead has differentiable and invertible transitions that

affect multiple post-decision states, then the Endogenous Grid Method can still be used. Consider a problem with two endogenous state variables and two post-decision states:

$$\begin{aligned} \mathrm{V}(x_1, x_2, s) &= \max_{a \in \Gamma(x_1, x_2, s)} \mathrm{W}(y_1, y_2, s) \\ &\quad s.t. \\ y_1 &= \mathrm{T}_1(x_1, a) \\ y_2 &= \mathrm{T}_2(x_2, a) \end{aligned} \tag{51}$$

where the continuation value is

$$\mathrm{W}(y_1, y_2, s) = \beta \mathbb{E}\left[\mathrm{V}'(\mathrm{G}_1(y_1, s), \mathrm{G}_2(y_2, s), s') | y_1, y_2, s\right] \tag{52}$$

Here, the first-order condition is

$$\frac{\partial \mathrm{W}(y_1, y_2, s)}{\partial y_1} \cdot \frac{\partial \mathrm{T}_1(x_1, a)}{\partial a} + \frac{\partial \mathrm{W}(y_1, y_2, s)}{\partial y_2} \cdot \frac{\partial \mathrm{T}_2(x_2, a)}{\partial a} = 0 \tag{53}$$

If T_2 *has the special structure* $\mathrm{T}_2(x_2, a) = x_2 + a + \mathrm{g}(a)$ *where* g *is differentiable with* $\frac{d\mathrm{g}}{da}$ *strictly monotone, and* $\frac{\partial \mathrm{T}_1}{\partial a}$ *is constant, then we can rearrange the first-order condition to obtain*

$$\frac{d\mathrm{g}}{da}(a) = -\left(\frac{\partial \mathrm{W}(y_1, y_2, s)}{\partial y_1} \Bigg/ \frac{\partial \mathrm{W}(y_1, y_2, s)}{\partial y_2}\right) \cdot \frac{\partial \mathrm{T}_1(x_1, a)}{\partial a} - 1 \tag{54}$$

Inverting yields the Endogenous Grid Method step:

$$a = \left(\frac{d\mathrm{g}}{da}\right)^{-1} \left(-\left[\frac{\partial \mathrm{W}(y_1, y_2, s)}{\partial y_1} \Bigg/ \frac{\partial \mathrm{W}(y_1, y_2, s)}{\partial y_2}\right] \cdot \frac{\partial \mathrm{T}_1(x_1, a)}{\partial a} - 1\right) \tag{55}$$

The strict monotonicity of $\frac{d\mathrm{g}}{da}$ *ensures existence and uniqueness of the inverse.[ĝ-monotone]*

6 Conclusion

We began by noting the tension between computational efficiency and economic realism in heterogeneous agent models. The Sequential Endogenous Grid Method resolves this tension through complementary contributions. **Conceptually**, we recognize that "simultaneous" from an economic perspective does not require "joint" from a computational perspective—separable structure lets us decompose multidecision problems into sequential stages that each admit an EGM inversion. **Computationally**, we develop interpolation methods (WGI for curvilinear grids, GPR for unstructured grids) that handle the non-rectilinear grids EGM generates, overcoming the bottleneck that has limited sequential approaches. Together, these contributions achieve the efficiency of EGM methods even in settings with multiple decisions and high-dimensional state spaces.

This paper introduces a novel method for solving dynamic stochastic optimization problems called the Sequential Endogenous Grid Method (EGM^n). Given a problem with multiple decisions (or control variables), the Sequential Endogenous Grid Method proposes separating the problem into a sequence of smaller subproblems that can be solved sequentially by using more than one EGM step. Then, depending on the resulting endogenous grid from each subproblem, this paper proposes different methods for interpolating functions on non-rectilinear grids, called the Warped Grid Interpolation (WGI) and the Gaussian Process Regression (GPR) method.

EGM^n is similar to the Nested Endogenous Grid Method (NEGM) of Druedahl [2021] and the Generalized Endogenous Grid Method (G2EGM) of Druedahl and Jørgensen [2017] in that it can solve problems with multiple decisions, but differs by exploiting full separability to chain multiple EGM inversions without any root-finding or optimization steps. This requires stronger structural assumptions than the triangular FOC systems of Iskhakov [2015] or the envelope methods of G2EGM, but when separability holds, eliminates computational overhead entirely. However, unlike G2EGM, which handles discrete choices and non-convexities through upper envelope techniques, EGM^n focuses on continuous choice problems where first-order conditions uniquely characterize optima. Extending EGM^n to handle non-convexities and discrete choices remains an important direction for future research. The contribution here is demonstrating how appropriate interpolation methods—WGI for curvilinear grids and GPR for unstructured grids—overcome computational bottlenecks that have limited the application of sequential EGM approaches, particularly when compared to Delaunay triangulation methods.

7 Appendix: Solving the illustrative G2EGM model with EGM^n

7.1 Problem Formulation

We designate as $\mathrm{w}_t(m_t)$ the problem of a retired household at time t with total resources m. The retired household solves a simple consumption-savings problem with no income uncertainty and a certain next period pension of $\underline{\theta}$.

$$\begin{aligned} \mathrm{w}_t(m_t) &= \max_{c_t} u(c_t) + \beta \mathrm{w}_{t+1}(m_{t+1}) \\ &\text{s.t.} \\ a_t &= m_t - c_t \\ m_{t+1} &= \mathsf{R}_a a_t + \underline{\theta}. \end{aligned} \tag{56}$$

Notice that there is no uncertainty and the household receives a retirement income $\underline{\theta}$ every period until death.

The value function of a worker household is

$$\mathrm{V}_t(m_t, n_t) = \mathbb{E}_\epsilon \max \left\{ v_t^0(m_t, n_t, \mathbb{W}) + \sigma_\epsilon \epsilon_{\mathbb{W}}, v_t^0(m_t, n_t, \mathbb{R}) + \sigma_\epsilon \epsilon_{\mathbb{R}} \right\} \tag{57}$$

where the choice specific problem for a working household that decides to continue working is

$$\begin{aligned} v_t^0(m_t, n_t, \mathbb{W}) &= \max_{c_t, d_t} u(c_t) - \alpha + \beta \mathbb{E}_t \left[\mathrm{V}_{t+1}(m_{t+1}, n_{t+1}) \right] \\ &\text{s.t.} \\ a_t &= m_t - c_t - d_t \\ b_t &= n_t + d_t + \mathrm{g}(d_t) \\ m_{t+1} &= \mathsf{R}_a a_t + \theta_{t+1} \\ n_{t+1} &= \mathsf{R}_b b_t \end{aligned} \tag{58}$$

and the choice specific problem for a working household that decides to retire is

$$v_t^0(m_t, n_t, \mathbb{R}) = \mathrm{w}_t(m_t + n_t). \tag{59}$$

7.2 Sequential Solution Method

The first step is to define a post-decision value function. Once the household decides their level of consumption and pension deposits, they are left with liquid assets they are saving for the future and illiquid assets in their pension account which they can't access again until retirement. The post-decision value function can be defined as

$$\begin{aligned} \mathfrak{v}_t(a_t, b_t) &= \beta \mathbb{E}_t \left[\mathrm{V}_{t+1}(m_{t+1}, n_{t+1}) \right] \\ &\text{s.t.} \\ m_{t+1} &= \mathsf{R}_a a_t + \theta_{t+1} \\ n_{t+1} &= \mathsf{R}_b b_t \end{aligned} \tag{60}$$

Then redefine the working agent's problem as

$$\begin{aligned} v_t^0(m_t, n_t, \mathbb{W}) &= \max_{c_t, d_t} u(c_t) - \alpha + \mathfrak{v}_t(a_t, b_t) \\ a_t &= m_t - c_t - d_t \\ b_t &= n_t + d_t + \mathrm{g}(d_t). \end{aligned} \tag{61}$$

Clearly, the structure of the problem remains the same, and this is the problem that G2EGM solves. We've only moved some of the stochastic mechanics out of the problem. Now, we can apply the sequential EGM^n method. Let the agent first decide d_t, the deposit amount into their retirement; we will call this the deposit problem, or outer loop. Thereafter, the agent will have net liquid assets of l_t and pension assets of b_t.

$$\begin{aligned} v_t^0(m_t, n_t, \mathbb{W}) &= \max_{d_t} v_t^1(l_t, b_t) \\ &\text{s.t.} \\ l_t &= m_t - d_t \\ b_t &= n_t + d_t + \mathrm{g}(d_t). \end{aligned} \tag{62}$$

Now, the agent can move on to picking their consumption and savings; we can call this the pure consumption problem or inner loop.

$$\begin{aligned} v_t^1(l_t, b_t) &= \max_{c_t} u(c_t) - \alpha + \mathfrak{v}_t(a_t, b_t) \\ &\text{s.t.} \\ a_t &= l_t - c_t. \end{aligned} \tag{63}$$

Because we've already made the pension decision, the amount of pension assets does not change in this loop and it just passes through to the post-decision value function.

Let's start with the pure consumption-saving problem, which we can summarize by substitution as

$$v_t^1(l_t, b_t) = \max_{c_t} u(c_t) - \alpha + \mathfrak{v}_t(l_t - c_t, b_t). \tag{64}$$

The first-order condition is

$$\frac{du}{dc}(c_t) = \frac{\partial \mathfrak{v}_t}{\partial a}(l_t - c_t, b_t) = \frac{\partial \mathfrak{v}_t}{\partial a}(a_t, b_t). \tag{65}$$

We can invert this Euler equation as in standard EGM to obtain the consumption function.

$$\mathfrak{c}_t(a_t, b_t) = \left(\frac{du}{dc}\right)^{-1}\left(\frac{\partial \mathfrak{v}_t}{\partial a}(a_t, b_t)\right). \tag{66}$$

Again as before, $\mathfrak{l}_t(a_t, b_t) = \mathfrak{c}_t(a_t, b_t) + a_t$. To sum up, using an exogenous grid of (a_t, b_t) we obtain the trio $(\mathfrak{c}_t(a_t, b_t), \mathfrak{l}_t(a_t, b_t), b_t)$ which provides an interpolating function for our optimal consumption decision rule over the (l, b) grid. Without loss of generality, assume $\mathfrak{l}_t = \mathfrak{l}_t(a_t, b_t)$ and define the interpolating function as

$$\check{c}_t(\mathfrak{l}_t, b_t) \equiv \mathfrak{c}_t(a_t, b_t). \tag{67}$$

For completeness, we derive the envelope conditions as well, and as we will see, these will be useful when solving the next section.

$$\begin{aligned} \frac{\partial v_t^1}{\partial l}(l_t, b_t) &= \frac{\partial \mathfrak{v}_t}{\partial a}(a_t, b_t) = \frac{du}{dc}(c_t) \\ \frac{\partial v_t^1}{\partial b}(l_t, b_t) &= \frac{\partial \mathfrak{v}_t}{\partial b}(a_t, b_t). \end{aligned} \tag{68}$$

Now, we can move on to solving the deposit problem, which we can also summarize as

$$v_t^0(m_t, n_t, \mathbb{W}) = \max_{d_t} v_t^1(m_t - d_t, n_t + d_t + \mathrm{g}(d_t)). \tag{69}$$

The first-order condition is

$$\frac{\partial v_t^1}{\partial l}(l_t, b_t)(-1) + \frac{\partial v_t^1}{\partial b}(l_t, b_t)(1 + \frac{d\mathrm{g}}{dd}(d_t)) = 0. \tag{70}$$

Rearranging this equation gives

$$\frac{d\mathrm{g}}{dd}(d_t) = \frac{\frac{\partial v_t^1}{\partial l}(l_t, b_t)}{\frac{\partial v_t^1}{\partial b}(l_t, b_t)} - 1. \tag{71}$$

Assuming that $\frac{d\mathrm{g}}{dd}(d)$ exists and is strictly monotone,[17] we can find

$$\mathfrak{d}_t(l_t, b_t) = \left(\frac{d\mathrm{g}}{dd}\right)^{-1} \left(\frac{\frac{\partial v_t^1}{\partial l}(l_t, b_t)}{\frac{\partial v_t^1}{\partial b}(l_t, b_t)} - 1\right) \tag{72}$$

Using this, we can back out n_t as

$$\mathfrak{n}_t(l_t, b_t) = b_t - \mathfrak{d}_t(l_t, b_t) - \mathrm{g}(\mathfrak{d}_t(l_t, b_t)) \tag{73}$$

and m_t as

$$\mathfrak{m}_t(l_t, b_t) = l_t + \mathfrak{d}_t(l_t, b_t). \tag{74}$$

In sum, given an exogenous grid (l_t, b_t) we obtain the triple $(\mathfrak{m}_t(l_t, b_t), \mathfrak{n}_t(l_t, b_t), \mathfrak{d}_t(l_t, b_t))$, which we can use to create an interpolator for the decision rule d_t.

To close the solution method, the envelope conditions are

$$\begin{aligned} \frac{\partial v_t^0}{\partial m}(m_t, n_t, \mathbb{W}) &= \frac{\partial v_t^1}{\partial l}(l_t, b_t) \\ \frac{\partial v_t^0}{\partial n}(m_t, n_t, \mathbb{W}) &= \frac{\partial v_t^1}{\partial b}(l_t, b_t). \end{aligned} \tag{75}$$

7.3 Supporting Calculations

To verify invertibility of g, note that the derivative $\frac{d\mathrm{g}}{dd}(d) = \frac{\chi}{1+d}$ is strictly positive for all $d > -1$ and $\chi > 0$, ensuring strict monotonicity. The inverse derivative is given by

$$\mathrm{g}(d) = \chi \log(1 + d) \qquad \frac{d\mathrm{g}}{dd}(d) = \frac{\chi}{1 + d} \qquad \left(\frac{d\mathrm{g}}{dd}\right)^{-1}(y) = \chi/y - 1. \tag{76}$$

The post-decision value and marginal value functions are

$$\begin{aligned} \mathfrak{v}_t(a, b) &= \beta \mathbb{E}_t \left[\mathrm{V}(m_{t+1}, n_{t+1})\right] \\ &\text{s.t.} \\ m_{t+1} &= \mathsf{R}_a a_t + \theta_{t+1} \\ n_{t+1} &= \mathsf{R}_b b_t \end{aligned} \tag{77}$$

[17] For $\mathrm{g}(d) = \chi \log(1 + d)$, we require $d > -1$ and $\chi > 0$ to ensure $\frac{d\mathrm{g}}{dd}(d) > 0$.

and

$$\begin{aligned}\frac{\partial \mathfrak{v}_t}{\partial a}(a_t, b_t) &= \beta \mathsf{R}_a \mathbb{E}_t \left[\frac{\partial \mathrm{V}_{t+1}}{\partial m}(m_{t+1}, n_{t+1}) \right] \\ &\text{s.t.} \\ m_{t+1} &= \mathsf{R}_a a_t + \theta_{t+1} \\ n_{t+1} &= \mathsf{R}_b b_t\end{aligned} \tag{78}$$

and

$$\begin{aligned}\frac{\partial \mathfrak{v}_t}{\partial b}(a_t, b_t) &= \beta \mathsf{R}_b \mathbb{E}_t \left[\frac{\partial \mathrm{V}_{t+1}}{\partial n}(m_{t+1}, n_{t+1}) \right] \\ &\text{s.t.} \\ m_{t+1} &= \mathsf{R}_a a_t + \theta_{t+1} \\ n_{t+1} &= \mathsf{R}_b b_t\end{aligned} \tag{79}$$

From discrete choice theory and the Discrete Choice Endogenous Grid Method (DCEGM) of Iskhakov et al. [2017], we know that

$$\mathbb{E}_t \left[\mathrm{V}_{t+1}(m_{t+1}, n_{t+1}, \epsilon_{t+1}) \right] = \sigma_\epsilon \log \left[\sum_{\mathbb{D} \in \{\mathbb{W}, \mathbb{R}\}} \exp \left(\frac{\mathrm{v}_{t+1}(m_{t+1}, n_{t+1}, \mathbb{D})}{\sigma_\epsilon} \right) \right] \tag{80}$$

and

$$\mathbb{P}_t(\mathbb{D} \mid m_{t+1}, n_{t+1}) = \frac{\exp \left(\mathrm{v}_{t+1}(m_{t+1}, n_{t+1}, \mathbb{D}) / \sigma_\epsilon \right)}{\sum_{\mathbb{D} \in \{\mathbb{W}, \mathbb{R}\}} \exp \left(\frac{\mathrm{v}_{t+1}(m_{t+1}, n_{t+1}, \mathbb{D})}{\sigma_\epsilon} \right)} \tag{81}$$

the first-order conditions are therefore

$$\frac{\partial \grave{\tilde{\mathfrak{v}}}_t}{\partial m}(m_{t+1}, n_{t+1}) = \sum_{\mathbb{D} \in \{\mathbb{W}, \mathbb{R}\}} \mathbb{P}_t(\mathbb{D} \mid m_{t+1}, n_{t+1}) \frac{\partial \mathrm{v}_{t+1}}{\partial m}(m_{t+1}, n_{t+1}, \mathbb{D}) \tag{82}$$

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